Bounding cross-trial temporal coupling in a loophole-free Bell test

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Abstract

Bell's theorem admits one loophole that no experiment can close in principle: measurement independence. Retrocausal and time-symmetric models exploit this opening, but the class of models in which the click probability at one trial depends on measurement settings at *other* trials has been formalised theoretically but never given an experimental bound. We bound it on public raw records from the CURBy device-independent randomness beacon — ten pulses of 1.5×10^7 trials each, spanning twenty-two months, analysed separately and never spliced.

The result that depends on no model is a bound in bits. Scanning every lag $|k| \leq 10{,}000$ between one party's outcome and the other party's setting,

I(O ; S at lag k) < 1.13×10^{-6} bits per trial,

at family-wise $\alpha = 0.05$ (Bonferroni, 40,002 hypotheses). This is 1/362 of the same-trial outcome–setting information measured on the same records, and the outcome–outcome information at nonzero lag is bounded below 1/13,384 of the Bell correlation at k = 0. As a guessing advantage, no outcome lets an adversary predict any other trial's setting with a bias above 1.25×10^{-3}.

Translated into the parameters of an explicit model — a click probability shifted by a kernel-weighted sum of other trials' settings, with kernel width τ and coupling strength ε — the same data exclude ε above 7.6×10^{-2} at $\tau = 1$ trial and 4.4×10^{-4} at $\tau = 10^4$ trials on a single pulse, for symmetric, future-only, past-only and one-sided exponential kernels: 208 tests, zero detections. This map is model-dependent by construction; it is the bound above re-expressed in the units of one model class, and within the tested class the sensitivity follows an approximately universal $1/\sqrt{Q}$ scaling with $Q = \sum_k W(k)^2$, the rigorous exclusion being stated for the four explicit families. Combining the ten pulses by inverse-variance weights, with the pulses shown to be uncorrelated, the excluded coupling falls to 1.9×10^{-2} and 1.2×10^{-4} at the same two widths: the strongest statement these data support, quoted alongside the single-pulse map rather than in its place.

The absence of cross-trial memory that device-independent randomness and DI-QKD assume is thereby checked empirically on an operating beacon, at the stated sensitivity. The result is null. Its value is that the region is now mapped rather than assumed empty.

1 Introduction

Bell's theorem rests on locality, realism, and measurement independence — the assumption that the settings chosen by the experimenters are statistically independent of whatever variables determine the outcomes. Loophole-free experiments [1,2,3] have closed the locality and detection loopholes. Measurement independence cannot be closed by experiment even in principle, since any test of it must presuppose that some choices are free [4,8].

It can nevertheless be *bounded*, and bounding it is not a philosophical exercise. Hall [4] showed that only a small relaxation of measurement independence suffices for a local model to reproduce quantum correlations, which makes the permitted size of the relaxation a physically meaningful quantity. Retrocausal programmes

[5,6,7] remain the principal route to a local account of Bell correlations, and are unexcluded by any experiment to date.

Two quantities must be kept apart. Hall's measure [4] is $I(X, Y : \Lambda)$, the mutual information between the settings and the hidden variable, and it admits no experimental upper bound in a standard Bell test [8]: Λ is by construction unobservable, and a model may place arbitrary correlation there. What we bound is $I(O; S)$ at nonzero lag, between two recorded quantities.

The two are related by the data-processing inequality in one direction only. Since the outcome is generated from λ, $I(O(i); S(i + k)) \leq I(\lambda(i); S(i + k))$; a small observed value does not force a small hidden one. The converse fails, and we do not claim it. The bound constrains models by their observable consequences, which is the only handle an experiment has, and is sufficient for the purpose: measurement dependence that produces no observable signature in the measured outcome statistics is not constrained by this analysis.

The retrocausal models cited above are almost exclusively *within-trial*: the outcome depends on the setting of the same measurement, chosen slightly later in time. This note addresses a different class — models in which the dependence extends *across* trials, so that the click probability at trial i is influenced by settings at trials $i \pm k$. Such models predict a directly observable signature: nonzero mutual information between an outcome and a setting separated in the trial sequence.

In the standard taxonomy of loopholes [25] the assumption at issue here is neither locality nor detection but the freedom-of-choice assumption, and specifically its cross-trial part: not whether a setting is correlated with the hidden variable of its own trial, but whether it is correlated with the outcome of another.

We are not aware of a direct experimental test of this class. The nearest neighbours are the memory loophole [10], where cross-trial dependence is admitted and handled by adversarially valid statistics rather than measured, and the theoretical treatment of measurement dependence across runs by Pope and Kay [14], which supplies no experimental limit. Published bounds on measurement dependence — the measurement-dependent locality parameter of Pütz et al. [15], bounded experimentally by Aktas et al. [16], and the spacetime-region bounds of the cosmic Bell tests [17,18] — are within-trial and in a different parametrisation, and are not directly comparable to ε. No published limit known to us constrains outcome–setting correlation as a function of trial separation.

Santos [26] asks a related question about the same physical effect and answers the opposite half of it. He shows that memory effects in photon-pair experiments cannot produce a loophole large enough to rescue local hidden variables — a statement about what memory could *accomplish*. What follows is a measurement of how much of it is *there*. Neither implies the other: a device could carry cross-trial memory far too weak to fake a Bell violation and still leak more than a protocol assuming none would tolerate. It is that quantity, at every separation up to ten thousand trials, that is measured here.

The nearest currency to ours is the one measured in bits. Barrett and Gisin [22] showed that if Bob's choices are completely free, every correlation obtainable from projective measurements on a singlet can be reproduced by a local model in which the mutual information between Alice's choice and the local variables is at most one bit — so a bound in bits on setting–variable correlation is the natural currency for how much measurement dependence a model needs. Hall and Branciard [23] sharpened the accounting in the CHSH scenario, finding that an optimal causal model needs about 0.080 bits of such information to reproduce the maximal quantum violation while a retrocausal one, in which later settings influence earlier source variables, needs about 0.046. Koh et al. [24] carried the same relaxation into device-independent randomness expansion and bounded what an adversary gains from it. All three quantify information involving the hidden variable, at a single trial; the bound below is on information between two recorded quantities, at a separation in trials, and the data-processing inequality relates the two in one direction only (above). It cannot be compared with them numerically, and we do not.

2 The model class

Notation. A *trial* is one entry of the beacon record. A *pulse* is one beacon round; the two words are used interchangeably below and rounds are identified by their beacon index. For trial i, $S_A(i), S_B(i) \in \{1, 2\}$ are the two parties' setting choices and $O_A(i), O_B(i) \in \{0, 1\}$ are their outcomes, 1 = detector click. Where the party is immaterial we write S and O. For the coupling term settings are encoded as $S(i) \in \{-1, +1\}$. The symbol k always denotes a lag in trials, never a count. The symbol λ denotes the hidden variable of the models under discussion; it is never a measured quantity in this work.

Measurement independence in its usual form asserts that the click probability at trial i is independent of all settings other than the local one at that trial. We relax this in a specific, parametrised way. Writing $p(i)$ for the click probability at trial i:

$$p(i) \;=\; p_0(S_{\text{own}}(i)) \;+\; \alpha\,\varepsilon \sum_k W_\tau(k)\, S_{\text{other}}(i+k) \tag{1}$$

Here $p_0(S_{\text{own}})$ is the ordinary same-trial click probability given the party's own setting — standard quantum mechanics, measured directly — and W_τ is a kernel of characteristic width τ normalised to max W = 1. Two parameters carry the physics:

- ε — coupling strength. We adopt the normalisation $\varepsilon = 1 \equiv$ as strong as the ordinary quantum-mechanical same-trial connection, so that ε is directly interpretable as a fraction of standard physics. This is a convention, not a physical statement. The factor α that implements it is defined in §3.
- τ — temporal extent, in trials. This is the parameter that gives the scan range physical meaning: without it, the choice of $|k| \leq 10{,}000$ would be arbitrary. §4.1 converts trials to seconds.

The scan range itself is not arbitrary either. Statistically it is almost free: a lag k is estimated from n − |k| trials, so at the edge of the window the sample is still 99.93% of the total and the loss of power is 0.03%; the sample would shrink by 10% only near $|k| \approx 1.5 \times 10^6$, where the loss of power would still be just 5%. In physical units the window is not a single number: at the instrument's nominal rate of about 250,000 trials per second (§4.1) it spans ≈ 40 ms, and the metadata bound of 51 μs per trial puts it at ≤ 510 ms. Only the shorter end can be assumed, so the coverage claim is made against 40 ms: electronic and fast mechanical correlation times of a table-top optical apparatus fall inside it, and anything slower — thermal drift of an optical table, which lives on seconds to minutes — does not. Slower mechanisms are not thereby unconstrained; they would appear as drift common to whole pulses rather than as structure in k, and are addressed by the setting-correlation test of §5.1 and by the ten-pulse comparison of §6.4, which spans twenty-two months.

Equation (1) is a linearisation, and holds while the perturbation is small against the baseline rate. Since the settings are independent and equiprobable, the injected term has standard deviation $\alpha\varepsilon\sqrt{Q}$ across trials, so the condition is $\alpha\varepsilon\sqrt{Q} \ll p_0$. The largest value of $\alpha\varepsilon\sqrt{Q}$ anywhere on the single-pulse map occurs for the past-only kernel at $\tau = 1$ in the channel O_B vs S_A, where $\varepsilon_{\text{excl}} = 1.382 \times 10^{-1}$ and Q = 0.386; with $\alpha_\text{B} = 1.8283 \times 10^{-3}$ this is 1.570×10^{-4}, or 2.3% of that channel's $p_0 = 6.87087 \times 10^{-3}$. Everywhere else it is smaller. The model is therefore used only in the regime where its linear form is accurate.

2.1 Why cross-trial coupling is a class worth testing

The model above is not introduced arbitrarily. Two established lines of work define hidden-variable classes with exactly this structure, and neither has been given an experimental bound as a function of separation.

The memory loophole. Barrett, Collins, Hardy, Kent and Popescu [10] observed that in a repeated Bell experiment the hidden variables of one trial may depend on the settings and outcomes of *previous* trials, since a real source and real detectors carry state between runs; the same reuse of stateful devices is what makes memory attacks possible in device-independent cryptography [19]. The loophole is normally addressed statistically, by deriving p-values valid without the i.i.d. assumption [11,12,13]. That approach establishes that a violation remains significant under adversarial memory; it does not measure how much memory is actually present. The present work asks the complementary question: given the data, how large could such a dependence be?

Measurement dependence across runs. Pope and Kay [14] formalised limited measurement dependence in multiple runs of a Bell test, treating the settings of one run as partially determined by variables shared with others. Their treatment is theoretical; it supplies no experimental limit. Our ε is a direct operational analogue of the quantity their framework leaves unconstrained.

Beyond these, three mechanisms would produce the signature we search for:

1. *Source memory.* A hidden variable retaining state across trials couples $\lambda(i)$ to conditions at neighbouring trials, including the settings applied there. The natural kernel width τ is then the correlation time of the source state in units of trials.

2. *Non-Markovian environment.* If the two wings share an environment with memory, the effective hidden variable at trial i is a functional of a window of the environment's history rather than of its instantaneous state.

3. *Temporally extended boundary conditions.* Retrocausal accounts [5,6,7] fix the hidden variable using a future boundary condition, and it matters what that condition is a condition *on*. In the terminology of [6] the settings are inputs to the model and the outcomes are outputs; such models are better called future-input dependent, and what they give up is λ-independence — the requirement that the distribution of λ be independent of the settings a and b that lie in its future. The future-dependence is therefore on the *setting*, a parameter fixed from outside the model. The outcome is not a second future condition; it is the thing the model is built to explain. What the scan of §3 measures is exactly the first quantity: $\delta(k)$ is the dependence of an outcome at trial i on the *setting* at trial $i+k$. Outcome–outcome dependence is a different object, bounded separately in §6.5, and is not what a future-input dependent account calls for.

   Such accounts are normally posed within a single trial, the relevant future input being the setting of the same measurement, and the condition is then imposed on an instantaneous hypersurface. That last step is the fragile one. In the two-boundary field models the weight from which probabilities are built is a flux integral of the field's four-current over a *closed* hypersurface in spacetime [20], and restricting that surface to two parallel instants is presented as a special case to be relaxed rather than as a requirement [21]. A duration enters those models explicitly: the predictions depend on the interval t_0 between the two boundaries, they depart from standard quantum mechanics if either boundary constrains the energy — a conserved quantity — to a precision near $\hbar/t_0$, and two measurements cannot be placed in a well-defined order at a separation shorter than the time a measurement itself takes [20].

   Our own reading of this, offered as an observation and not as anyone's stated position, is that the durationless instant is the part that does not survive. A future condition posed physically is carried by something that persists over an interval, so the "moment of measurement" acquires a finite extent, and a constraint of finite extent reaches neighbouring trials. τ is that extent, in units of trials, and the future-only kernels of §2.2 are its natural form.

   The formalisation of this picture remains open. We know of no model that derives a cross-trial kernel W(k) from a two-boundary or future-input dependent starting point, and we do not supply one. The bound does not require it: the only property of the kernel that enters the exclusion is $Q(\tau) = \sum_k W(k)^2$ (§2.2, §6.3), so whatever form the formalisation takes, its coupling is bounded by the tables of §6.3 as soon as its W is evaluated on the same lag grid.

A fourth possibility is mundane and worth stating, because the test excludes it too: correlated drift between the setting generators and the detector state would produce cross-trial outcome–setting correlation with no new physics whatsoever. A null result is therefore also a systematics check on the experiment, independent of any interpretation.

What is bounded is a rate, not a hidden variable. We model the click probability p(i), an observable, and bound the mutual information between two observables. This is deliberate and is the reason the bound exists at all. A dependence residing entirely in an unobservable λ, leaving no trace in the outcome statistics, is not constrained here — but neither does it do any work: a measurement dependence that never reaches the outcomes cannot help a local model reproduce quantum correlations. What we bound is the operationally effective part.

2.2 Kernels

Four kernels are considered. In every one of them the lag k = 0 is excluded: $W_\tau(0) = 0$ by definition. The k = 0 term would be the dependence of an outcome on the setting of the *same* trial — the no-signalling condition, a different physical question from cross-trial coupling, already tested separately in the "Cross-pair I at k = 0" row of Table 3 (§5). A model class named cross-trial must not contain the within-trial term, and the matched filter of §6.3 must not integrate over it. Every kernel, Q, and filter in this paper is therefore evaluated over $0 < |k| \leq 10{,}000$.[1]

The quantity

$$Q(\tau) \;=\; \sum_{0<|k|\leq 10{,}000} W_\tau(k)^2 \tag{2}$$

is the only property of the kernel that enters the final bound (§6.3), so it is tabulated here alongside each shape:

[1] An earlier version of this analysis included k = 0 in the symmetric kernel. Removing it changes the symmetric bounds at small τ only — by a factor of 1.7 at $\tau = 1$, 3–7% at $\tau = 10$, about 1% at $\tau = 100$ and below 0.5% for $\tau \geq 1{,}000$ — and leaves the one-sided kernels, which never contained k = 0, unchanged.

Table 1: The four kernel shapes and their Q relative to the symmetric kernel.

Kernel	W(k)	$Q(\tau)/Q_{\mathrm{sym}}$, $\tau \geq 10$	at $\tau = 1$
symmetric	$\exp(-k^2/2\tau^2)$, all $k \neq 0$	1	1
future-only	$\exp(-k^2/2\tau^2)$ for $k > 0$, else 0	0.500	0.500
past-only	$\exp(-k^2/2\tau^2)$ for $k < 0$, else 0	0.500	0.500
exponential future	$\exp(-k/\tau)$ for $k > 0$, else 0	≈ 0.28	0.203

With k = 0 absent from all four, the one-sided Gaussians carry exactly half the symmetric Q at every τ, to machine precision. The exponential ratio is not a constant: for $\tau \gtrsim 10$ it lies between 0.270 and 0.289 (0.270 at $\tau = 10$, 0.281 at $\tau = 100$, 0.282 at $\tau = 1{,}000$, 0.289 at $\tau = 10^4$, where the window truncates the two kernels differently), which is why the column is given as ≈ 0.28. Q is computed exactly as a finite sum over the scan window, never asymptotically; the symmetric kernel has Q = 0.773 at $\tau = 1$ and 2.545 at $\tau = 2$. One detail of the discrete lag grid: since k = 0 is excluded, every kernel peaks at k = ±1, where W is $\exp(-1/2\tau^2)$ or $\exp(-1/\tau)$ rather than exactly 1 — 0.61 and 0.37 at $\tau = 1$, within 1% of 1 for $\tau \geq 10$. The analysis uses these W as written. At small τ the quoted ε is therefore expressed in units of this sub-unity peak; restated in strictly peak-normalised units the excluded coupling would be *smaller* by the same factor, so the tables err on the conservative side.

The one-sided kernels matter because retrocausal models are future-input dependent [6] and therefore asymmetric; a symmetric-only analysis would leave the natural case untested. The sign convention is $k > 0 \equiv$ future: the setting at trial i+k influences the outcome at trial i.

A consequence of excluding k = 0 deserves note. None of the four filters touches the single lag that carries genuine quantum correlation and known systematics; the nearest lags, k = ±1, are where detector deadtime would appear, and Table 3 shows it does not at this sample size. What the filters integrate is cross-trial structure only.

3 Observable signature

Write p_0 for the baseline click probability and $\delta(k)$ for half the difference in click rate between the two settings at lag k. The model of §2 gives

$$\delta(k) \;=\; \alpha\, \varepsilon\, W_\tau(k) \tag{3}$$

where α is the device sensitivity. For small δ, standard expansion of the mutual information about independence yields

$$I(k) \;\approx\; \frac{\delta(k)^2}{2\ln 2\; p_0(1-p_0)} \;=\; C\,\varepsilon^2 W_\tau(k)^2, \qquad C = \frac{\alpha^2}{2\ln 2\; p_0(1-p_0)} \tag{4}$$

For the Gaussian kernel this is a bell curve of width $\tau/\sqrt{2}$ in I and τ in δ.

Here p_0 is the *marginal* click probability of the 2×2 table, $p_0 = (c_1 + c_2)/n$ with c_s the click count under setting s and n the number of trials — not an average taken as an approximation. Because the settings are balanced 50/50 this coincides with $(r_1 + r_2)/2$ to within 4.9×10^{-5} relative, the residual being $(m_1 - m_2)(r_1 - r_2)/2n$ with m_s the number of trials under setting s and $r_s = c_s/m_s$. The marginal is the expansion point at which the second-order form is exact to leading order; the alternative choices r_1, r_2, or $\sqrt{r_1 r_2}$ would give ratios of 1.37, 0.77, and 1.03 against the measured value, whereas the marginal gives 0.986.

Calibration. α is not a free parameter. It is fixed by the measured same-pair lag-0 dependence, which is ordinary quantum mechanics. On round 28297 the two settings give click rates $r_1 = 4.96662 \times 10^{-3}$ and $r_2 = 8.89072 \times 10^{-3}$, hence $\delta(0) = 1.96206 \times 10^{-3} \pm 2.14 \times 10^{-5}$ and $\alpha = 1.9621 \times 10^{-3} \pm 1.1\%$. The same construction on Bob's wing gives $r_1 = 5.04249 \times 10^{-3}$, $r_2 = 8.69908 \times 10^{-3}$ and $\alpha_{\mathrm{B}} = 1.8283 \times 10^{-3}$, with marginal click probabilities $p_0 = 6.92833 \times 10^{-3}$ for Alice and 6.87087×10^{-3} for Bob. Both are needed: the map is computed per channel, and every quantity quoted for O_{B} vs S_{A} — the dashed curves of Figure 3, the joint bounds of Table 7, and the worst point of the map — uses α_{B}, not α_{A}.

The second-order approximation reproduces the exactly computed mutual information to 1.35% (4.036×10^{-4} against 4.091×10^{-4}, the latter from the four-term $p \log p$ sum with no expansion). The relevant small parameter is not p_0 but $\delta/p_0 = 0.28$. The prediction is low, hence conservative. This gives C = 4.036×10^{-4} bits per trial per ε^2.

The exclusion map does not use this approximation. The bound of §6.3 is built from $\hat{\varepsilon} = T/(\alpha Q)$ with $T = \sum_k W(k)\hat{\delta}(k)$ from measured click rates, $\alpha = \delta(0) = (r_2 - r_1)/2$, and $Q = \sum_k W(k)^2$ purely geometric. None of these contains p_0 or any expansion. Reconstructing the map from these raw quantities and comparing against the published values at 208 points gives a maximum relative difference of 2.2×10^{-16}. The approximation enters only the model-independent bound of §6.1 and the translation constant C.

The two settings differ in click rate by a factor 1.79, an intentional asymmetry of the Eberhard configuration [9]. The resulting large α works in our favour: since $\varepsilon_{\mathrm{excl}} \propto 1/\alpha$, a device sensitive to its settings yields a tight bound on ε.

Verification. The derivation was tested rather than assumed. Synthetic coupling of known (ε, τ) was injected into the real data — real settings retained, so that the true setting autocorrelation is preserved — and the resulting $I(k)$ compared against prediction. Across τ = 1, 10, 100, 1000 and three ε each, mean ratios were 1.012 (δ amplitude), 1.002 (δ width), and 1.005 (I width against the predicted $\tau/\sqrt{2}$), the fits taken over k ≠ 0 since the kernel has no k = 0 term (§2.2). An $\varepsilon = 0$ control was run before each τ and produced a flat curve in every case. Injected probabilities are clipped to [0,1]; a point is declared invalid if more than 0.1% of trials clip, and the ε of each injection is chosen automatically to stay below that limit. The observed clipping never exceeded 0.026%.

4 Data

The CURBy beacon [3] publishes per-trial records from a loophole-free Bell test at NIST: settings and outcomes for both parties, in acquisition order, prior to randomness extraction. Each pulse contains 1.5×10^7 trials after the protocol stopping criterion.

Ten pulses were analysed: five consecutive (rounds 28293–28297, spanning 65 minutes on 2025-08-22) and five spread across the archive (rounds 1000, 15000, 22000, 23000, 26000; 2023-10-31 to 2025-06-28). The full span is twenty-two months. These are reported separately, since pulses from different dates necessarily differ in derived calibration parameters. The protocol configuration is identical across all ten, verified over 25 rounds spanning the archive. The click rate falls from 0.93% (round 1000) to 0.64% (round 26000), with the 2025 pulses at 0.69%: the apparatus changed materially across the interval sampled.

Exact download URLs, byte counts and SHA-256 digests for every round used are recorded in `results/curby_manifest.json` of the code repository, so the input can be verified bit for bit.

4.1 Trials to seconds — an upper bound only

The beacon metadata contains no per-trial timing field. What it does contain is stage timestamps (`request` → `precommit` → `randomness`). The interval `request -> precommit` covers acquisition *plus* unknown overhead, so it yields an **upper bound**: for round 28297, 774.34 s over 15,190,485 recorded trials, i.e. ≤ **51.0** μ**s per trial**. On that bound $\tau = 1$ is ≤ 51 μs and $\tau = 10{,}000$ is ≤ 510 ms.

This is not a constant of the dataset. For the ten pulses analysed here:

Table 2: Upper bound on the time per trial, for each analysed pulse. The record count is constant to 0.3%; the acquisition interval is not.

Round	Date	request → precommit	Records	μs per trial (upper bound)
1000	2023-10-31	168.4 s	15,228,817	11.1
15000	2024-03-07	350.9 s	15,227,680	23.0
22000	2024-07-17	483.3 s	15,223,405	31.7
23000	2025-05-13	510.5 s	15,182,270	33.6
26000	2025-06-28	598.7 s	15,191,281	39.4
28293	2025-08-22	641.4 s	15,191,016	42.2
28294	2025-08-22	647.0 s	15,182,571	42.6
28295	2025-08-22	679.4 s	15,186,766	44.7
28296	2025-08-22	730.7 s	15,190,447	48.1
28297	2025-08-22	774.3 s	15,190,485	51.0

The bound grows by a factor of 4.6 in twenty-two months at constant record count, for reasons the metadata does not explain. The published description of the instrument [3] states a nominal rate — "about 250,000 trials every second", with 15 million trials acquired in ≈ 60 s — for the same protocol configuration analysed here (the 15,000,000-trial stopping criterion). At that nominal rate one trial is ≈ 4 μs, $\tau = 1$ is ≈ 4 μs and $\tau = 10{,}000$ is ≈ 40 ms, and the request → precommit interval is dominated by overhead rather than acquisition, which would also account for its growth across the archive. The nominal figure is quoted alongside the metadata bound, not in place of it: it is the only per-trial timing the records themselves support, and the nominal rate is not verifiable from the records. **The τ axis is therefore reported in trials throughout. The conversion above applies to round 28297 only, is an upper bound rather than a measurement, and must not be transferred to other rounds.**

5 Validation

No result was accepted before the following passed. Significance for mutual-information quantities is quoted throughout as $\sqrt{G}$, where $G = 2n \ln 2 \cdot I$ is asymptotically $\chi^2(1)$ for a 2×2 table (§6.1). Two entries use the Eberhard statistic instead and are marked: there the scale is $\mathrm{J}/\sigma_{\mathrm{J}}$, defined in §6.2.

Table 3: Validation checks passed before any result was accepted.

Check	Expected	Observed
Eberhard J at k = 0	violation	+11.7 to +18.8 (J/σ_J), all ten pulses
Same-pair I at k = 0	large	4.091×10^{-4} bits, G = 8,508 → **92.2σ ($\sqrt{G}$)**
Cross-pair I at k = 0	zero (no-signalling)	2.14×10^{-8} bits (0.67σ) and 2.09×10^{-11} bits (0.02σ)
Shift sign	injected k = +3 recovered at k = +3	+134.2 (J/σ_J) at k = +3, on synthetic data
Detector deadtime	visible at |k| = 1, absent beyond	**2.7σ ($\sqrt{G}$) at k = -1 in O_B vs S_B — observed but not significant**
Mirror filter	one-sided filters distinguish direction	correct filter +10, mirrored 1.56
Setting independence	zero at every lag	max 4.83σ over 20,001 lags, 0 above threshold (§5.1)
Pulse independence	$\hat{\delta}(k)$ uncorrelated between pulses	max |r| = 0.018 over 90 pulse pairs (2.6σ), 0 above 3σ (§6.4)
Oscillatory coupling	no periodic structure in $\hat{\delta}(k)$	0 of 200,000 frequencies above threshold, ten pulses (§6.8)

The deadtime row is a negative finding and is reported as such. The largest effect at k = -1 anywhere in round 28297 is 2.7σ ($\sqrt{G}$) in Bob's same-pair channel; Bob's cross-pair channel gives 2.4σ and both of Alice's channels give below 0.6σ. All are far below the 4.85σ threshold of §6.1, so the experiment does *not* resolve detector deadtime at this sample size, and the argument for instrument sensitivity rests on the same-trial correlation (362× the bound) and the Bell correlation (13,384× the bound) instead. An earlier draft quoted "+6.2σ" for this row; that figure came from a 100-shuffle empirical null whose standard deviation was 25% below the $\chi^2(1)$ value, on a different significance scale, and is withdrawn.

The shift-sign test is not decorative: an error of sign would exchange "future" and "past", inverting the question. The mirror-filter test establishes that the one-sided filters are genuinely directional rather than symmetric filters in disguise.

Three implementation errors were found and corrected during development: a bit extraction returning $\{0,3\}$ rather than $\{0,1\}$; a block-reversal permutation producing out-of-range indices for lengths that are not powers of two; and an invalid null comparing biased data against unbiased surrogates. Each would have invalidated everything downstream. They are reported here because the credibility of a null result rests entirely on the auditing that produced it.

5.1 Correlated settings would fake the signal

One mechanism would produce exactly the signature we search for without any new physics, and it must be excluded before the bound means anything. Alice's outcome depends strongly on Alice's *own* setting — that is ordinary quantum mechanics, and it is the large effect this analysis calibrates against, $\delta_A = 1.96 \times 10^{-3}$. If Bob's setting at trial i+k were correlated with Alice's at trial i, the cross-pair channel would show a signal built entirely from that one dependence.

The size is fixed by the algebra. With settings encoded as ±1 and equiprobable, $\mathbb{E}[O_A \mid S_B(i+k) = s] = p_0 + \delta_A\, \rho(k)\, s$ with $\rho(k) = \mathrm{Corr}(S_A(i), S_B(i+k))$, so

$$\delta_{\text{phantom}}(k) \;=\; \delta_A \,\rho(k) \tag{5}$$

and, passed through the same matched filter, the fake coupling is

$$\varepsilon_{\text{phantom}}(\tau) \;=\; \frac{1}{Q(\tau)} \sum_k W_\tau(k)\,\rho(k) \tag{6}$$

in which α cancels: the phantom ε is just the kernel-weighted mean of the setting correlation, directly comparable with $\varepsilon_{\text{excl}}$.

Measuring $\rho(k)$ for every $|k| \le 10{,}000$ by FFT:

Table 4: Correlation between the two setting sequences, and each sequence with itself, across the full lag range. The standard error is $1/\sqrt{n} = 2.58 \times 10^{-4}$, evaluated at the analysed sample of n = 15,000,000 trials — the stopping-criterion cut described in the footnote of §6.1. At the full record count of the file, 15,190,485, it would be 2.566×10^{-4}, 0.6% smaller.

Correlation	max $\lvert\rho\rvert$	at lag	in units of $1/\sqrt{n}$	above threshold
$\mathrm{Corr}(S_A(i), S_B(i+k))$	1.246×10^{-3}	$-9,438$	4.83σ	0 / 20,001
$\mathrm{Corr}(S_A(i), S_A(i+k)),\ k \neq 0$	9.683×10^{-4}	−635	3.75σ	0 / 20,001
$\mathrm{Corr}(S_B(i), S_B(i+k)),\ k \neq 0$	1.104×10^{-3}	+9,301	4.28σ	0 / 20,001

Nothing reaches the 4.848σ threshold, and the same conclusion follows from the mutual information of the settings computed with the machinery and threshold of §6.1: max $I(S_A(i); S_B(i+k)) = 1.121 \times 10^{-6}$ bits per trial at the same lag, G = 23.29 against a threshold of 23.50, 0 of 20,001 above.

The largest point deserves its own sentence, because it is the closest call in this work. Across the 20,001 lags the standardised correlations are textbook normal — mean −0.004, standard deviation 1.005, Kolmogorov–Smirnov p = 0.86, 62 lags above 3σ against 54 expected — and the 4.83σ point at lag $-9,438$ is isolated, its immediate neighbours being 1.15, 0.14, -0.24 and -0.27σ. Its family-corrected p-value is 0.027: a one-in-forty fluctuation among twenty thousand, which is what a Bonferroni threshold exists to absorb. It is also at a lag unrelated to anything in the outcome analysis.

Taken at face value regardless, the phantom signal is far below the sensitivity. The largest δ_{phantom} is 2.45×10^{-6}, which is 0.11 of the statistical error on $\hat{\delta}$ at a single lag (2.14×10^{-5}). Through the matched filter, across all four kernels and all 26 values of τ, $\varepsilon_{\text{phantom}}$ never exceeds 0.0073 of $\varepsilon_{\text{excl}}$: **the fake signal is at least 138 times below the single-pulse bound at every point of the map**.

All ten pulses. Since the headline result is the joint bound of §6.4, a correlated pair of generators in *any* pulse would contaminate it. The same test was therefore run on all ten:

Table 5: Setting cross-correlation in each of the ten pulses. Nothing anywhere reaches the threshold, by correlation or by mutual information.

Round	max $|\rho|$	at lag	in units of $1/\sqrt{n}$	above threshold
1000	9.761×10^{-4}	+4,532	3.78σ	0 / 20,001
15000	1.144×10^{-3}	+1,277	4.43σ	0 / 20,001
22000	1.176×10^{-3}	+2,137	4.56σ	0 / 20,001
23000	1.085×10^{-3}	$-2,286$	4.20σ	0 / 20,001
26000	1.088×10^{-3}	+1,106	4.21σ	0 / 20,001
28293	1.077×10^{-3}	$-4,930$	4.17σ	0 / 20,001
28294	1.230×10^{-3}	+9,968	4.76σ	0 / 20,001
28295	1.029×10^{-3}	+9,439	3.98σ	0 / 20,001
28296	1.096×10^{-3}	+8,518	4.25σ	0 / 20,001
28297	1.246×10^{-3}	$-9,438$	4.83σ	0 / 20,001

Nothing in any pulse reaches the threshold, on either statistic: 0 of 200,010 tests by correlation and 0 of 200,010 by mutual information. Seen across the whole set, the 4.83σ point discussed above also loses its edge of interest: the expected largest |z| among 200,010 standard normals is 4.56, so observing 4.83 has a family-corrected p of 0.24. It is the ordinary maximum of a large sample, not a feature.

Against the joint bound the phantom is correspondingly small. Taking the worst single pulse against the joint bound gives a ratio of 0.039; propagating the phantom into the joint estimate with the same inverse-variance weights as the data gives 0.015. **The fake signal is at least 65 times below the joint bound at every point of the map.** Setting independence, to the precision the data allow, is not an assumption here but a measurement — in every pulse used.

6 Results

6.1 Model-independent bound

Full scan of round 28297, 20,001 lags per pair, 40,002 hypotheses in total. The empirical Bonferroni threshold is impractical at this multiplicity (it would require ~8×10^5 shuffles), so we use the analytic result that $G = 2n \ln 2 \cdot I$ is asymptotically $\chi^2(1)$ for a 2×2 table. This was calibrated against 2,000 real shuffles per pair before use: mean G = 0.988 and 0.995 against 1.000 predicted, KS p = 0.59 and 0.69.

Confidence level. All bounds in this paper are stated at family-wise $\alpha = 0.05$ with a Bonferroni correction over 40,002 hypotheses, i.e. per-test p = 1.25×10^{-6}, equivalently $z > 4.848$ on the $\sqrt{G}$ scale. The corresponding mutual-information threshold is 1.130×10^{-6} bits per trial.[2]

[2] All statistics in this paper are computed on exactly the first 15,000,000 records of each pulse — the protocol stopping criterion, the same cut the beacon's own extraction pipeline applies. The raw files carry 15,182,270 to 15,228,817 records (§4.1), so 1.20% to 1.50% of each pulse is discarded. Evaluated at the full 15,190,485 records of round 28297, the threshold would be 1.116×10^{-6} bits per trial, 1.3% lower; truncation can only cost sensitivity, so the stated numbers are conservative.

Table 6: Largest outcome–setting mutual information over all 20,001 lags, round 28297.

Pair	max I	at lag	% of threshold	above threshold
O_A vs S_B	6.985×10^{-7}	$-3,194$	62%	0 / 20,001
O_B vs S_A	7.108×10^{-7}	$-9,992$	63%	0 / 20,001

$$\boxed{I\big(O; S \text{ at lag } k\big) \ < \ 1.13 \times 10^{-6} \text{ bits per trial} \quad \text{for every } |k| \leq 10{,}000, \ \ \alpha_{\mathrm{FW}} = 0.05} \tag{7}$$

This is 1/362 of the same-trial value. Bonferroni does not assume independence; neighbouring lags are correlated, which renders the correction conservative. Family-wise control is the appropriate criterion here, and not a false-discovery rate such as Benjamini–Hochberg: an FDR procedure controls the expected fraction of false positives *among the discoveries*, which with zero discoveries constrains nothing, whereas an upper bound is by definition a statement about the family-wise probability of any false detection. The positive correlation of neighbouring lags makes Bonferroni conservative for that purpose, as already noted. The full scan is shown in **Figure 1**.

Guessing advantage. The same bound reads directly as a limit on prediction. Consider an adversary who tries to guess the setting S_B(i+k), a uniformly random bit, from Alice's outcome O_A(i) at some lag k. For a uniform binary S the advantage of the optimal guess equals the total-variation distance between the two conditional outcome distributions, and Pinsker's inequality applied to the joint distribution gives

$$\left|2P(\hat{S} = S) - 1\right| \ \leq \ \sqrt{2 \ln 2 \ I(O; S)} \tag{8}$$

with I in bits. At I < 1.13×10^{-6} bits the bias is below 1.25×10^{-3}, so $P(\hat{S} = S) < 0.5 + \text{bias}/2 = 0.50063$, at the family-wise confidence above. The constant was checked rather than copied: the binary form is the tight one for a uniform bit, an exact inversion of the binary entropy through Fano's inequality, $H(S|O) \leq h(P_{\mathrm{err}})$, gives 1.2516×10^{-3} as well, agreeing to seven digits, and the inequality was verified numerically over a grid of binary channels. This is the quantity a leakage lemma would take as input; we do not carry the argument through to a min-entropy statement, and the bound is stated per lag, not jointly over lags. What it means for device-independent protocols is discussed in §6.7.

6.2 The $J(k)$ curve

The Eberhard statistic [9] for the CH-type inequality used by this apparatus is

$$J \ = \ N(11|a_1 b_1) - N(10|a_1 b_2) - N(01|a_2 b_1) - N(11|a_2 b_2) \ \leq \ 0 \tag{9}$$

under local realism, where $N(o_A o_B \mid a\, b)$ counts trials with the given settings and outcomes, and its uncertainty is the Poisson propagation $\sigma_J = \sqrt{N_1 + N_2 + N_3 + N_4}$ over the four counts entering J. Round 28297 gives J = 1,816 with σ_J = 154.9, i.e. **+11.7 (J/σ_J)**.

Computing the same statistic with settings from trial i and outcomes from trial i+k gives, for round 28297, +11.7 at $k = 0$ and a mean of −135.5 over the 106 nonzero lags (range −137.3 to −133.2) — not merely consistent with zero, but far below the violation boundary. Unlike the mutual-information scan of §6.1, which is exhaustive over all 20,001 lags, J(k) is evaluated at 107 sampled lags: every lag from −50 to +50, together with ±100, ±1,000 and ±10,000 — 106 nonzero lags plus the k = 0 control. The reason is cost, not principle:

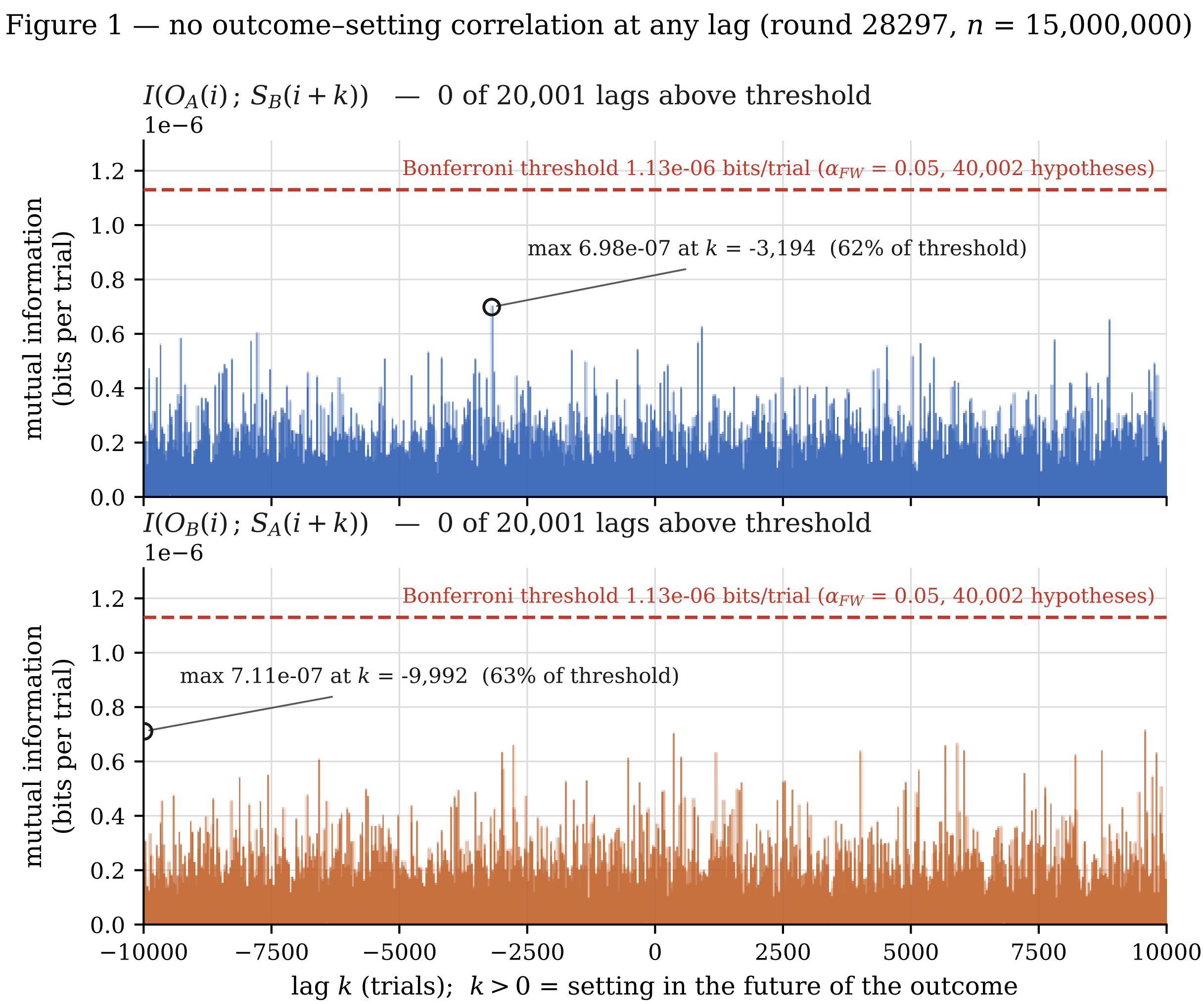


Figure 1: Mutual information between one party's outcome at trial i and the other party's setting at trial i+k, for all 20,001 lags $|k| \leq 10{,}000$, round 28297. Both cross-pair channels are shown. The dashed line is the Bonferroni threshold at family-wise $\alpha = 0.05$ over 40,002 hypotheses. The largest value in either channel reaches 63% of the threshold. Vector version: `figures/fig1_mi_vs_lag.pdf`.

each J(k) requires a separate pass over the full record, since the statistic has no FFT shortcut, and as a sanity check rather than a bound it does not need exhaustive coverage. The algebra is direct: when settings and outcomes decouple, all four setting groups see the same outcome distribution, so the two $N(11|\cdot)$ terms cancel in expectation and the two surviving single-click terms, which enter negatively, dominate. The baseline level is pulse-dependent: across the ten pulses the least negative nonzero lag ranges from −133.2 (round 28297) to −168.6 (round 1000).

This is a sanity check, not a statistical test. By the identity of Appendix C.2 the decoupled value of J is large and negative by construction, so agreement with zero is not the question and no p-value is attached to it. The question is whether J ever becomes *positive* — whether a violation appears where none can exist. Across all ten pulses, $k \neq 0$ with $J > 0$: **0 of 1,060** (106 nonzero lags × 10 pulses). The peak falls from +11.7 to the baseline within a single trial. There is no temporal tail. See **Figure 2**.

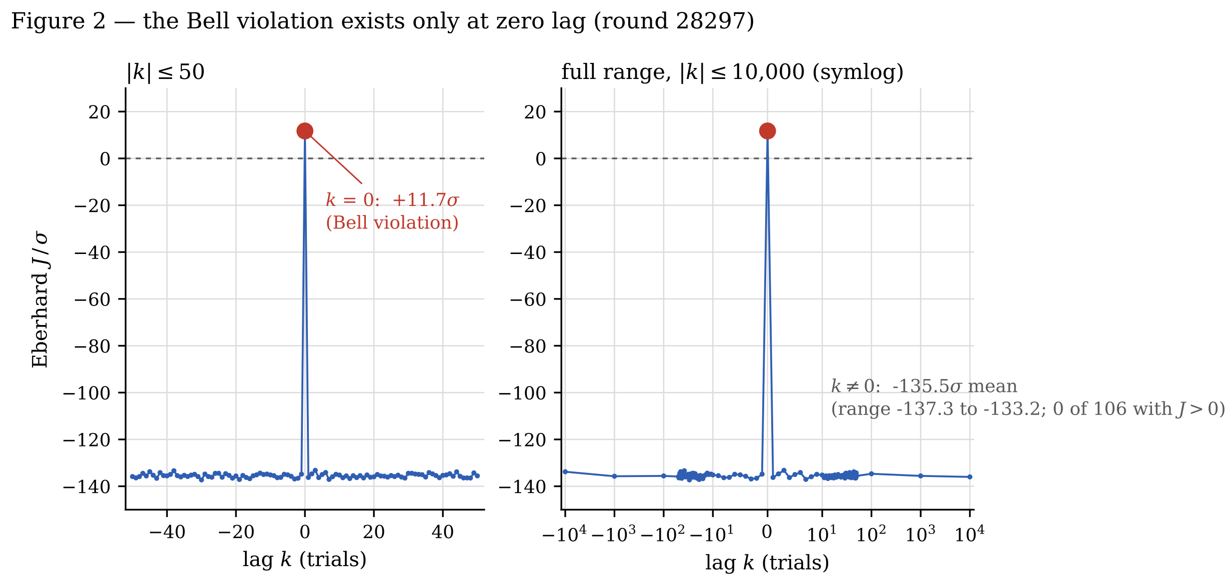


Figure 2: Eberhard J in units of its Poisson error, as a function of the lag between settings and outcomes, round 28297. Left: |k| ≤ 50, linear. Right: the full scanned range on a symmetric-log axis, showing that the behaviour at |k| up to 10,000 is identical to that at $|k| = 1$. The violation is confined to exactly one lag. Vector version: `figures/fig2_j_curve.pdf`.

6.3 Exclusion map

The map of this section is not a second result. It is the bound of §6.1 translated into the units of the model class of §2: the same measured click-rate differences $\hat{\delta}(k)$, weighted by an assumed kernel and converted to a coupling strength through the measured α. Whatever one thinks of the kernels, the bits of §6.1 stand on their own; what the map adds is a reading of those bits in the language a model would use.

The statistic. Let $\hat{\delta}(k)$ be the estimator of $\delta(k)$ from the data: half the measured difference in click rate between the two settings at lag k, computed from the 2×2 table at that lag. The optimal statistic is a matched filter on $\hat{\delta}$ rather than on $\hat{I}$: $\hat{\delta}$ is linear in the signal with noise symmetric about zero, whereas $\hat{I}$ is quadratic with positive-definite noise that the filter would accumulate as bias. Define

$$T(\tau) \;=\; \sum_k W_\tau(k)\,\hat{\delta}(k), \qquad \mathbb{E}[T] = \alpha\,\varepsilon\,Q(\tau), \qquad \hat{\varepsilon} = \frac{T}{\alpha Q}, \qquad z = \frac{T}{\sigma_T} \tag{10}$$

Explicitly, $\hat{\varepsilon}_A = T_A/(\alpha_A Q)$ for the pair O_A vs S_B, and $\hat{\varepsilon}_B = T_B/(\alpha_B Q)$ for O_B vs S_A: each channel is converted to a coupling strength with the sensitivity of the party whose *outcomes* are being scanned — α_A for Alice's outcomes, even though the settings entering the filter are Bob's. Here σ_T is the standard deviation of T under the null. σ_T is obtained two ways and **the larger of the two is used**: analytically, $\sigma_T = \sqrt{\sum_k W(k)^2 \sigma_\delta(k)^2}$ from the binomial error of each 2×2 table, and empirically, from 400 shuffles of the setting sequence per pair. The empirical and analytic values agree to within 1.8% on average per kernel (0.94–1.06 per individual τ). Two assumptions were verified rather than assumed: the $\hat{\delta}(k)$ are uncorrelated across k (|r| ≤ 0.014 at $\Delta k = 1, 2, 3, 5, 10$, against a standard error of 0.007), and the null is centred (mean z under shuffling ≤ 0.05 in magnitude).

Optimality, and what it assumes. A matched filter is the Neyman–Pearson optimal statistic for a signal of *known shape* in *Gaussian white* noise. Both conditions are testable on these data and both were tested. Whiteness is the uncorrelatedness of $\hat{\delta}(k)$ across k already quoted, |r| ≤ 0.014. Gaussianity: standardising each lag by its own binomial error, the 20,001 values $\hat{\delta}(k)/\sigma_\delta(k)$ have mean +0.0007 and standard deviation 0.9984 with a Kolmogorov–Smirnov p of 0.535 for O_A vs S_B, and mean +0.0062, standard deviation 1.0000, p = 0.223 for O_B vs S_A. The tails, which are what a 4.85σ threshold actually depends on, match too: 922 and 884 lags beyond 2σ against 910 expected, 49 and 65 beyond 3σ against 54, none beyond 4σ against 1.3 expected. Where the shape is *not* known — the case the $1/\sqrt{Q}$ regularity below bears on — optimality is lost and the filter is merely valid: the threshold and the bound remain correct, but some other filter could be more sensitive.

Threshold. The same $z > 4.848$ as §6.1 is used, i.e. the Bonferroni correction for 40,002 hypotheses is *borrowed* for a family of 208 tests. The matched correction for 208 tests would be $z > 3.672$; under it, every bound on the map would tighten by 16% to 24% (mean 22%). The borrowed threshold is therefore deliberately over-conservative by design, and keeps the map directly comparable with the model-independent bound.

Grid. τ takes 26 values: `round(logspace(0, 4, 25))` together with 30 and 300, i.e. 1, 2, 3, 5, 7, 10, 15, 22, 30, 32, 46, 68, 100, 147, 215, 300, 316, 464, 681, 1000, 1468, 2154, 3162, 4642, 6813, 10000.

$\varepsilon_\mathrm{excl}(\tau) = |\hat{\varepsilon}| + z_\mathrm{thr}\, \sigma_T/(\alpha Q)$, for O_A vs S_B:

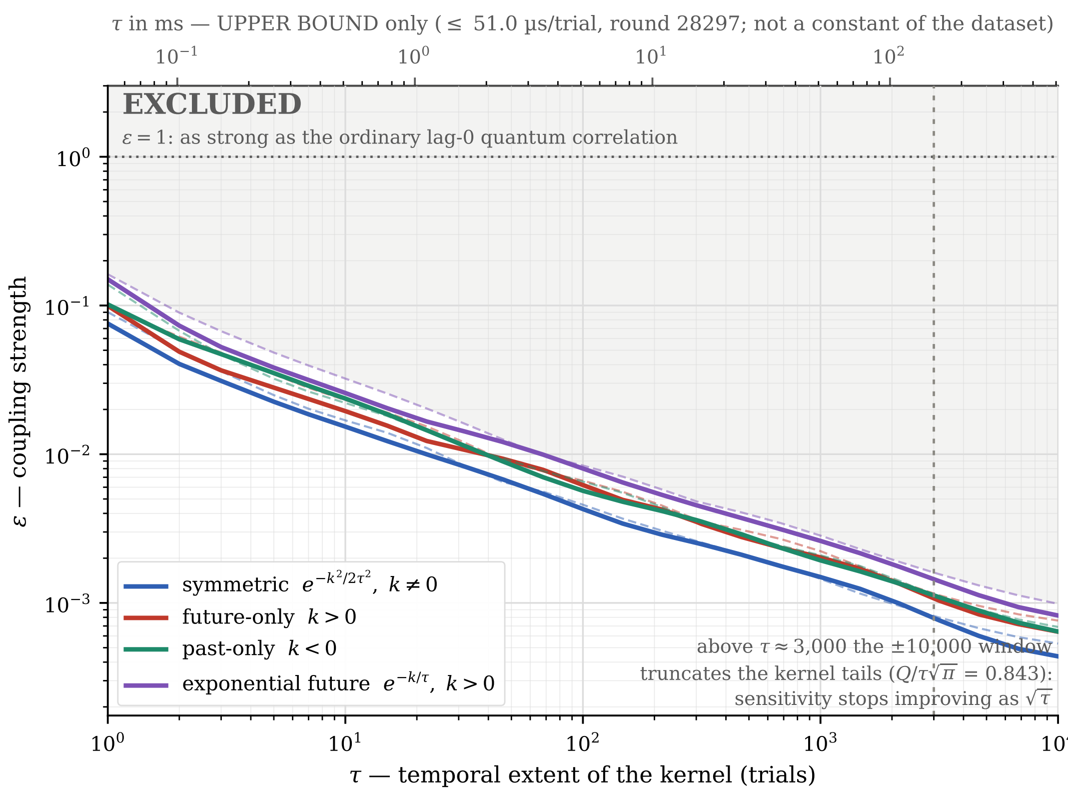
Table 7: Excluded coupling strength $\varepsilon_\mathrm{excl}(\tau)$ for the four kernels, O_A vs S_B.

τ (trials)	symmetric	future-only	past-only	exp. future
1	7.559×10^{-2}	9.954×10^{-2}	1.017×10^{-1}	1.502×10^{-1}
10	1.533×10^{-2}	1.949×10^{-2}	2.368×10^{-2}	2.580×10^{-2}
100	4.277×10^{-3}	6.207×10^{-3}	5.674×10^{-3}	8.014×10^{-3}
1,000	1.493×10^{-3}	2.023×10^{-3}	1.933×10^{-3}	2.613×10^{-3}
10,000	4.373×10^{-4}	6.407×10^{-4}	6.420×10^{-4}	8.230×10^{-4}

Zero detections in 208 tests (4 kernels × 26 values of τ× 2 pairs). Maximum $|z| = 2.36$. The map is shown in **Figure 3**.

The one-sided bounds relax by the geometrically expected factor: predicted $\sqrt{2} = 1.414$ against a measured median of 1.38 (range 1.18–1.56) for the one-sided Gaussians, and predicted $\sqrt{1/0.282} = 1.88$ against a median of 1.85 (1.66–1.99) for the exponential, over both pairs and $\tau \geq 3$. With k = 0 absent from every kernel the expected factor is the same at all τ; the scatter about it is the random data term $|\hat{\varepsilon}|$, which is largest at small τ, where each filter rests on a handful of lags. These are factors of order unity, not orders of magnitude. The worst value anywhere on the map is 1.62×10^{-1} — even at its weakest point, the analysis excludes coupling above one sixth of ordinary quantum mechanics.

Figure 3 — exclusion map: coupling strength ε versus kernel width τ, four kernel shapes



Solid: O_A vs S_B. Dashed: O_B vs S_A. Shaded region excluded at family-wise α = 0.05 (Bonferroni, 40,002 hypotheses, $z > 4.848$).

Figure 3: Upper bound on the coupling strength ε as a function of kernel width τ, for all four kernel shapes and both cross-pair channels, round 28297. The shaded region is excluded at family-wise $\alpha = 0.05$. The bend above $\tau \approx 3{,}000$ is the finite scan window truncating the kernel tails. The upper axis gives the upper-bound conversion of §4.1 and applies to this round only. Vector version: `figures/fig3_exclusion_map.pdf`.

Generalisation beyond the four shapes. Since the noise on $\hat{\delta}(k)$ is approximately constant in k, the bound depends on the kernel only through $Q(\tau) = \sum_k W(k)^2$, and $\varepsilon_{\text{excl}} \propto 1/\sqrt{Q}$. This was verified numerically on the threshold term of the bound, $z_{\text{thr}}\sigma_{\text{T}}/(\alpha\text{Q})$, which is the part that carries the kernel dependence: its product with $\sqrt{Q}$ has a standard deviation of 3.5% across all 208 points, most of which is the difference between the two pairs ($\alpha_A \neq \alpha_B$), with within-kernel scatter of about 2%. The full product $\varepsilon_{\text{excl}} \cdot \sqrt{Q}$ scatters more (8%), because it also contains the random data term $|\hat{\varepsilon}|$ of each kernel; the constant below is taken from the full product, so that scatter is included in c. The tested class is explicit: four families × 26 widths, 104 shapes, each measured in both channels. For pre-specified kernels within the tested class, the observed sensitivity follows an approximately universal $1/\sqrt{Q}$ scaling; the rigorous exclusion is stated for the four explicit families. The table below records that empirical regularity, tabulated by Q, and is offered as a reading aid — not as a theorem, and not as an exclusion of shapes that were not tested:

Table 8: The empirical $1/\sqrt{Q}$ regularity of the bound within the tested class, tabulated by Q. An empirical regularity, not a theorem; the rigorous exclusion is the one stated for the four explicit families.

Q	$\varepsilon_{\text{excl}}$	example shape
3	4.07×10^{-2}	narrow kernel over ~3 lags
10	2.23×10^{-2}	Gaussian $\tau \approx 5.6$, exponential $\tau \approx 20$
100	7.05×10^{-3}	Gaussian $\tau \approx 56$, square over 100 lags
1,000	2.23×10^{-3}	Gaussian $\tau \approx 564$, exponential $\tau \approx$ 2,000
10,000	7.05×10^{-4}	Gaussian $\tau \approx 5{,}642$

with $\varepsilon_{\text{excl}}(Q) = c/\sqrt{Q}$, c = 0.0705 taken as the maximum of $\varepsilon \cdot \sqrt{Q}$ over the 168 points with $\tau \geq 10$ (mean 0.0598), so that the table never promises a tighter bound than was measured; checked against all four measured kernels at τ = 10, 100, 1,000 and 10,000, the formula is looser than the measurement by factors of 1.03 to 1.32. Three conditions apply: W normalised to max W = 1; summation within $|k| \leq 10{,}000$; and a kernel specified in advance whose weight varies smoothly over its support. The last condition is not decorative. The data term $|\hat{\varepsilon}|$ of each kernel is a random draw of the noise with standard deviation $0.011/\sqrt{Q}$, of which c covers about 1.3 standard deviations on average — ample for kernels that average over many lags the way the four tested families do, but a kernel concentrated on a few isolated lags can align with individual fluctuations of $\hat{\delta}(k)$ and exceed the tabulated value: an indicator kernel placed on the three largest same-sign $\hat{\delta}$ of this dataset would need 6.8×10^{-2}, 1.7 times the Q = 3 entry. Such concentrated kernels — and everything with Q < 3 — should be read against the measured $\hat{\delta}(k)$ directly rather than from the table.

This addresses the arbitrariness of the Gaussian choice only in part. Within the tested class the sensitivity is set by Q and not by the details of the shape, which is why the Gaussian was not a special choice; but a kernel outside the tested class is not thereby excluded, and no claim is made for it.

The map was verified by injection at four levels — $\varepsilon = 0$, $\varepsilon_{\text{excl}}/2$, $\varepsilon_{\text{excl}}$ and $2\varepsilon_{\text{excl}}$ — with ten repetitions per point, at two values of τ for each of the three asymmetric kernels: six points, sixty injections at each level. Detection is 0/60 at $\varepsilon = 0$, 2/60 at half the bound, 39/60 at the bound and 60/60 at twice it, and the mean z scales linearly with ε (**Figure 4**). The 39/60 at the bound itself is expected: a confidence bound has by construction approximately 50% power at the bound, and quoting anything else would misrepresent it.

Kernel mismatch. The mirror-filter test covers the extreme case, a filter pointed the wrong way. The intermediate case is a filter of the wrong *width*. Injecting a symmetric Gaussian signal of width τ_{true} and filtering it with a range of τ_{filter} gives, as a fraction of the matched value:

Figure 4 — a signal at the stated bound is recovered; twice the bound is always recovered

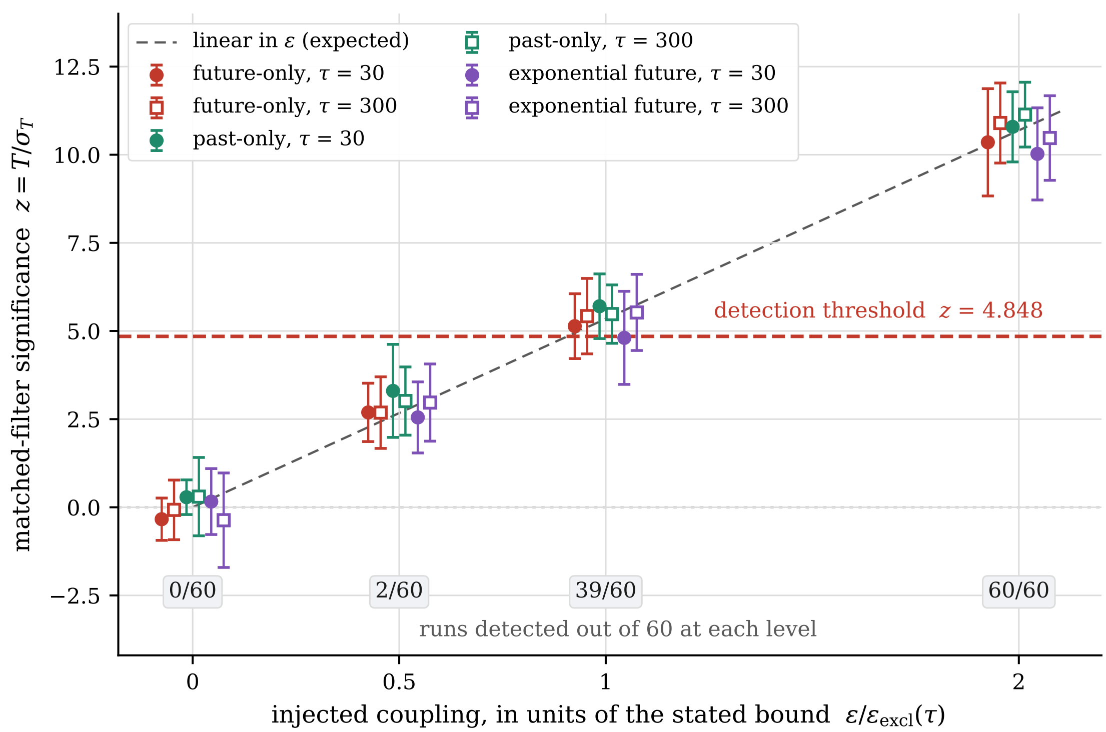


Figure 4: Recovery of an injected signal of known strength. The horizontal axis is the injected coupling in units of the bound this analysis states for that kernel and τ; the vertical axis is the matched-filter significance. Points are the mean over ten repetitions, error bars their standard deviation, and the six series are the three asymmetric kernels at $\tau = 30$ and $\tau = 300$. The dashed line is the linear scaling expected if the filter is unbiased. The boxes give the number of runs detected out of 60 at each level. Nothing is found when nothing is injected, and a signal at twice the stated bound is found every time. Vector version: `figures/fig4_injection_power.pdf`.

Table 9: Loss of significance when the filter width does not match the signal width, measured over five injections at $4\varepsilon_{\rm excl}$ and compared with the closed form $\sum_k W_f(k)W_t(k)/\sqrt{\sum_k W_f(k)^2 \cdot \sum_k W_t(k)^2}$, which is 1 when the widths match. The last column departs from the closed form at $\tau_{\rm true} = 3{,}000$ because the $\pm 10,000$ window truncates a filter of width 30,000.

$\tau_{\rm filter}/\tau_{\rm true}$	0.1	1/3	0.5	0.825	1	1.21	2	3	10
$\tau_{\rm true} =$ 30	0.425	0.777	0.897	0.990	1	0.993	0.904	0.784	0.440
$\tau_{\rm true} =$ 300	0.434	0.794	0.909	0.995	1	0.987	0.882	0.758	0.425
$\tau_{\rm true} =$ 3,000	0.433	0.769	0.887	0.989	1	0.992	0.903	0.822	0.734
closed form	0.444	0.774	0.894	0.991	1	0.991	0.894	0.774	0.445

The τ grid is logarithmic with step $10^{4/24} = 1.468$, so a signal falling exactly between two grid points is mismatched by a factor 1.212 — and loses **0.9%** of z. The grid is dense enough that nothing can hide between its points. Even gross misspecification degrades gracefully: a factor of two costs 10%, a factor of three 22%, a factor of ten 56%. The filter is never blind, only less sensitive.

Measured z initially exceeded prediction by 3–7%. The cause is not physical: the map uses $\sigma_{\rm T}$ = max(empirical, analytic) whereas the verification computes z from the analytic value alone. With 400 shuffles the empirical estimate carries ±3.5% noise, and the max() systematically selects upward fluctuations. The measured ratio $z_{\rm obs}/z_{\rm pred} = 1.035$ decomposes as $\sigma_T(\text{map})/\sigma_T(\text{analytic}) = 1.018$ times a residual of 1.016 ± 0.015, consistent with unity. Doubling the shuffle count from 400 to 800 moves the ratio from 1.046 to 1.007 as expected for an estimation artefact. An alternative mechanism — reduced autocorrelation in injected data — was tested and excluded: the outcome autocorrelation is zero at every lag from 1 to 5 (maximum $|r| = 3.1 \times 10^{-4}$ against a standard error of 2.6×10^{-4}), and $\sigma_{\rm T}$ computed from shuffles of injected versus real data agrees to within error with the sign varying between cases.

Sensitivity ceases to improve as $\sqrt{\tau}$ above $\tau \approx 3{,}000$, where the $\pm 10,000$ scan window begins truncating the kernel tails ($Q(10^4)/\tau\sqrt{\pi} = 0.843$). This is visible as a bend in Figure 3 and is not smoothed.

6.4 Joint bound from the ten pulses

The map above uses one pulse. The other nine can be combined with it, but not by concatenation: they are separated by months, the click rate falls by a third across them, and α varies by 20% (Limitation 1). Splicing them into a single sequence would create an object with no physical counterpart and would import the drift as spurious structure.

Each pulse is therefore analysed on its own terms — its own $\hat{\delta}(k)$, its own $\sigma_{\rm T}$ from 400 shuffles of its own setting sequence, its own α — and the resulting estimates are combined with inverse-variance weights:

$$\hat{\varepsilon}_{\rm joint} = \frac{\sum_p \hat{\varepsilon}_p/\sigma_p^2}{\sum_p 1/\sigma_p^2}, \qquad \sigma_{\rm joint} = \Big(\sum_p 1/\sigma_p^2\Big)^{-1/2} \tag{11}$$

with $\hat{\varepsilon}_p = T_p/(\alpha_p Q)$ and $\sigma_p = \sigma_{T,p}/(\alpha_p Q)$, and the joint bound formed as before, $|\hat{\varepsilon}_{\rm joint}| + z_{\rm thr}\sigma_{\rm joint}$. A pulse with small α has large $\sigma_{\rm p}$ and contributes less: sensitivity sets the weight, not trial count.

Table 10: Joint exclusion from all ten pulses, O_A vs S_B, with the improvement over the single-pulse map of §6.3.

τ (trials)	symmetric	future-only	past-only	exp. future	gain over §6.3
1	1.945×10^{-2}	2.655×10^{-2}	2.564×10^{-2}	4.345×10^{-2}	3.4–3.8
10	3.564×10^{-3}	5.898×10^{-3}	5.783×10^{-3}	8.380×10^{-3}	3.1–4.4
100	1.145×10^{-3}	1.625×10^{-3}	1.653×10^{-3}	2.339×10^{-3}	3.4–3.8
1,000	3.523×10^{-4}	4.976×10^{-4}	4.940×10^{-4}	7.077×10^{-4}	3.8–4.3
10,000	1.234×10^{-4}	2.089×10^{-4}	2.188×10^{-4}	2.770×10^{-4}	3.0–3.6

Zero detections in 208 joint tests, maximum $|z_{\text{joint}}| = 2.10$. The mean improvement over the single-pulse map is a factor **3.62** (range 2.77 to 5.10), which exceeds the naive $\sqrt{10} = 3.16$ for the reason given in Limitation 1: round 28297 has the third-smallest α of the ten, within 2% of the smallest, so the comparison is against one of the least sensitive pulses. The gain column is computed against the independent re-estimate of the 28297 map described below, so that both sides of the ratio carry the same shuffle noise; dividing the printed tables instead reproduces it to within the ±3.5% noise of the empirical σ_T. The worst joint bound anywhere on the map is 4.34×10^{-2}, against 1.62×10^{-1} for the single pulse.

As a by-product this run re-estimates the 28297 map independently, with a different shuffle seed; it reproduces the published values of §6.3 with a mean ratio of 1.005 and a range of 0.95 to 1.08, consistent with the ±3.5% estimation noise on the empirical σ_T discussed above.

Independence of the pulses. Inverse-variance weighting is optimal, and σ_{joint} is correct, only if the ten estimates are uncorrelated. Five of the pulses (28293–28297) were acquired consecutively within 65 minutes, so this cannot be assumed. Since T_p is a linear functional of the $\hat{\delta}(k)$ of pulse p, any covariance between pulses would have to appear as a correlation between the $\hat{\delta}(k)$ of one pulse and the $\hat{\delta}(k)$ of another across lags. This was measured directly: for each of the 45 pairs of pulses and each channel, the Pearson correlation of the standardised $\hat{\delta}(k)/\sigma_\delta(k)$ of one pulse with that of the other over the 20,000 lags k ≠ 0, whose standard error under independence is $1/\sqrt{20{,}000} = 0.0071$. The largest |r| among the 90 values is 0.018 (2.6σ, pulses 23000 and 28293, which are three months apart); among the twenty values from the five consecutive pulses the largest is 0.015 (2.1σ, pulses 28294 and 28295), and the mean is −0.002. The standard deviation of r across pairs is 0.0072 and 0.0077 in the two channels, against 0.0071 expected, and no pair exceeds 3σ. The pulses are uncorrelated at the level the data can resolve, and the diagonal covariance assumed by the weights is measured rather than assumed. Had a substantial correlation appeared, the combination would have required generalised least squares with the full covariance matrix; it does not.

That test is at the level of the lags. The quantity the joint bound actually depends on is the covariance of the estimators T_p themselves, and covariance concentrated in the few lags where a narrow kernel puts its weight would move an unweighted correlation over 20,000 lags very little while still biasing σ_{joint}. It was therefore computed directly, for each of the 104 filters, each of the 45 pulse pairs and both channels. Since $T_p = \sum_k W(k)\,\hat{\delta}_p(k)$ and the $\hat{\delta}$ are uncorrelated across k, the covariance is $\sum_k W(k)^2 \mathrm{Cov}(\hat{\delta}_p, \hat{\delta}_q)$, whose estimator standardised by its own sampling error is $t_{pq} = \sum_k W(k)^2 z_p z_q / \sqrt{\sum_k W(k)^4}$ with $z = \hat{\delta}/\sigma_\delta$, standard normal if the pulses are independent. A W^2-weighted *correlation* would be the wrong statistic and would saturate at ±1: the effective number of lags a filter uses, $(\sum W^2)^2/\sum W^4$, is 1.1 for the future kernel at $\tau = 1$. Across the 9,360 values of t the mean is +0.029 and the standard deviation 0.955 against 0 and 1, and the largest is 4.18 where 4.28 is expected as the maximum of that many normals. Keeping the measured off-diagonal terms would move σ_{joint} by +1.3% on average over the filters that rest on ten or more lags; against a null built by giving each pulse an independent random circular shift of its lag axis, which destroys any alignment between pulses while preserving each pulse's own structure, that shift is +0.58σ, and the corresponding figure over all filters is +0.86σ. There is no covariance beyond what independent pulses produce, and the diagonal weighting stands.

The combination also assumes a coupling common to the ten pulses — not a triviality across twenty-two months of drifting hardware. The assumption was tested rather than left implicit: for each kernel and τ, the heterogeneity statistic $Q_{\text{het}} = \sum_p (\hat{\varepsilon}_p - \hat{\varepsilon}_{\text{joint}})^2/\sigma_p^2$ follows $\chi^2(9)$ if one ε underlies all ten pulses. Over the 208 points of the map its mean is 8.0 against the expected 9.0 — mildly under-dispersed, the signature of the conservative σ_{T} = max(empirical, analytic) — and the largest value, 28.5, has a family-corrected p of 0.17 across the 208 strongly correlated points. The ten pulses are statistically consistent with a single coupling, here zero. If the coupling nevertheless varied across epochs, the joint bound would constrain its inverse-variance-weighted mean, and the per-pulse maps (§6.3, Limitation 1) remain the epoch-resolved statements.

The joint bound is the strongest statement this dataset supports. The single-pulse map of §6.3 is retained as the quoted result elsewhere in this paper because §6.1, §6.2, §6.5 and the injection verification are all single-pulse and are directly comparable with it.

6.5 Outcome–outcome correlation

If the trial sequence were temporally scrambled — Alice's and Bob's records mispaired — the signature would appear not in outcome–setting but in outcome–outcome correlation. Scanning I(O_{A}(i) ; O_{B}(i+k)):

Table 11: Outcome–outcome scan, round 28297.

k = 0 (positive control)	1.4267×10^{-2} bits, G = 296,670 (545σ on the $\sqrt{G}$ scale)
max at k ≠ 0	7.3232×10^{-7} bits, at lag $-2,526$
above threshold	0 / 20,000

$$\boxed{I\big(O_A(i); O_B(i+k)\big) \ < \ 1.066 \times 10^{-6} \text{ bits per trial} \quad \text{for every } 0 < |k| \le 10{,}000} \tag{12}$$

This test involves a single pair rather than two, so the Bonferroni family is 20,001 hypotheses rather than 40,002 and the threshold is correspondingly lower, 1.066×10^{-6} instead of 1.130×10^{-6}. The count "0 / 20,000" excludes k = 0, which is the positive control and is expected to exceed the threshold — the 20,001st hypothesis is the control itself.

The bound is 1/13,384 of the Bell correlation at k = 0. Detector deadtime does not appear here either (G ≤ 1.61, i.e. ≤ 1.3σ, at $|k| \le 2$).

Repeated identically on all ten pulses, the scan gives **0 of 200,000** nonzero lags above threshold, the largest value being 4.4σ; the k = 0 control lies between 508σ and 552σ in every pulse, and the coincidence enhancement over independent outcomes grows from 38× (round 1000) to 59× (the 2025 pulses) as the click rate falls.

6.6 Single-party memory

The scans above are cross-party: one party's outcome against the *other* party's setting. Memory of a device's *own* settings — I(O_{A}(i) ; S_{A}(i+k)) at k ≠ 0 — is a distinct channel. It cannot fake a Bell violation by itself, but it is the natural signature of a stateful detector or setting generator, and until now it had been examined only at $|k| = 1$, as the deadtime entry of §5, on one pulse. The same 20,001-lag scan as §6.1 — same $\chi^2(1)$

machinery, same threshold — was therefore run on both same-party channels, O_A vs S_A and O_B vs S_B, in all ten pulses. The lag k = 0 is the positive control: it carries the ordinary same-trial setting dependence that this analysis calibrates against ($\sqrt{G}$ between 85.8 and 139.5 across the twenty channels) and is excluded from the count.

Zero detections in 400,000 tests (20,000 nonzero lags × 2 channels × 10 pulses). The largest value anywhere is $\sqrt{G}$ = 4.82, at lag $-5,819$ of round 1000 (O_B vs S_B), reaching 98.8% of the threshold — and this is what the null predicts for a family of this size: the expected number of $\chi^2(1)$ draws above that point in 400,000 tests is 0.58. Neither device retains a measurable memory of its own settings at any lag the window reaches, in any pulse.

6.7 Parity of setting pairs

Every test so far is marginal in the settings: it asks whether the outcome depends on *one* setting at *one* lag. A dependence on a joint function of several settings whose single-setting marginals vanish — the simplest being the parity of two adjacent settings — would be invisible to all of it. The parity of a balanced independent pair is itself a balanced binary sequence, so the identical machinery applies: define P(i) = 1 if S(i) = S(i+1), and scan I(O_A(i) ; P_B(i+k)) and I(O_B(i) ; P_A(i+k)) over all 20,001 lags, in all ten pulses, at the same threshold. No lag is a positive control here: the parity is independent of each single setting by construction, so even k = 0 must be null. The $\chi^2(1)$ calibration was re-verified on the parity sequence directly (mean G = 1.025 over 200 shuffles).

Zero detections in 400,020 tests (20,001 lags × 2 channels × 10 pulses). The largest value anywhere is $\sqrt{G}$ = 4.50, at lag $-1,777$ of round 28297, reaching 86% of the threshold. The outcomes depend on no adjacent-pair parity anywhere in the window. This closes the quadratic-adjacent case only; functionals of three or more settings, and of non-adjacent pairs, remain untested (Limitation 7).

Relevance to device-independent protocols. Device-independent randomness generation and device-independent quantum key distribution rest on the assumption that the devices carry no exploitable memory between trials — that the statistics of trial i are not conditioned on the settings of other trials. That assumption is exactly the memory loophole [10], and it is the reason such protocols are analysed with adversarially valid, memory-tolerant statistics [11–13] rather than with the i.i.d. estimates that would otherwise suffice. Dropping the assumption costs more than a looser bound: Weilenmann, Budroni and Navascués [27] show that in the non-i.i.d. regime whole classes of certification become impossible, membership tests for non-convex sets of correlations among them, so how well the assumption holds decides what can be certified at all. The results of this paper — 0 of 400,000 single-party memory tests (§6.6), 0 of 200,010 setting-correlation tests (§5.1), and the bound of 1.13×10^{-6} bits per trial on the information any outcome carries about any other trial's setting (§6.1) — constitute a direct empirical check of that assumption on an operating beacon, and the check is independent of any interpretation of the model class of §2. It does not establish the security of the protocol: what is verified is one assumption, at the stated sensitivity, on these records.

6.8 Oscillatory kernels: a frequency scan

All four kernel families of §2.2 are one-signed, and an oscillatory coupling is nearly orthogonal to every one of them: it would pass the matched filters of §6.3 unseen. It is also the shape with the most mundane possible origin — mains pickup, a modulation of the pump, any periodic drive in the apparatus — which makes it worth testing rather than conceding.

The statistic. A Lomb–Scargle periodogram of the standardised $\hat{\delta}(k)/\sigma_\delta(k)$ over the 20,000 lags k ≠ 0. Lomb–Scargle rather than a plain periodogram because the lag grid is uniform apart from the single missing point at k = 0, which it handles exactly instead of by interpolation. In the Scargle normalisation the power at each frequency is Exp(1) under Gaussian white noise, and both conditions were established in §6.3: the $\hat{\delta}(k)$ are uncorrelated across k (|r| ≤ 0.014) and Gaussian (Kolmogorov–Smirnov p = 0.535 and 0.223). The frequency grid is $f_j = j/20{,}001$ cycles per lag for j = 1 . . . 10,000, so M = 10,000 searched frequencies per channel, and the Bonferroni threshold on Exp(1) is $z = \ln(m/0.05)$: z = 12.90 for the two channels of one pulse (m = 20,000) and z = 15.20 for the ten pulses (m = 200,000), the latter matching the family construction of §6.5, §6.6 and §6.7. Frequencies are reported in cycles per lag; the conversion to Hz is nominal, through the ~4 μs per trial of §4.1, and the metadata upper bound of 51 μs per trial would divide every frequency by 12.7.

Round 28297. In O_A vs S_B the largest power is 9.02, at a period of 2.4 lags, against an expected null maximum of $\ln M + \gamma = 9.79$: nothing. In O_B vs S_A the largest power is 13.38, at f = 0.00330 cycles per lag, a period of 303 lags — 825 Hz on the nominal conversion, 65 Hz on the metadata bound, neither of which is a mains harmonic. That value sits just above the single-pulse threshold, at 104% of it, and below the ten-pulse threshold. Its probability within one channel is $1 - (1 - e^{-P})^M = 0.015$, so roughly 0.03 across the two channels of the pulse: a one-in-thirty fluctuation, which is what a threshold set at $\alpha = 0.05$ is built to admit. The mean power is 1.0000 in both channels, exactly the Exp(1) expectation, and the frequency nearest 50 Hz nominal carries power 1.20 and 0.78.

It does not repeat. A periodic drive in the apparatus would appear in more than one record, and above all in the five pulses taken consecutively within 65 minutes of one day. The other nine pulses were therefore scanned in full, both channels, and the 303-lag frequency examined in each. No channel of any of them has a maximum above even the single-pulse threshold — the largest anywhere is 12.22 — so **no frequency in the ten pulses reaches the ten-pulse threshold: 0 of 200,000**. At the 303-lag frequency itself the power across the other eighteen pulse-channels has mean 1.12 and maximum 3.86, an unremarkable Exp(1) sample; in the other channel of round 28297 it is 0.53. The excess is confined to one channel of one pulse and is consistent with noise. It is reported rather than dropped, and the ten-pulse scan is reported as what it is: a follow-up run after the single-pulse scan produced it, not an independent pre-specified test.

Sensitivity and control. A sinusoid of amplitude A in white noise of width σ carries Lomb–Scargle power $\approx NA^2/4\sigma^2$, so the threshold corresponds to a coupling $\varepsilon_\mathrm{thr} = \sqrt{4z/N}\,\sigma_\delta/\alpha = 5.5 \times 10^{-4}$ for an oscillatory kernel of unit peak — comparable to the $\varepsilon_\mathrm{excl}$ the one-signed kernels reach near $\tau \approx 10^3$, and the number that now replaces the untested caveat of Limitation 2. The machinery was verified by injecting an oscillatory coupling at the outcome level, $\lambda(i) = \lambda_0 + \alpha\varepsilon \sum_{k\neq 0} \cos(2\pi f_0 k + \varphi)\, S_B(i + k)$, with a fresh random phase each repetition: at 2ε_thr the peak is found, and found at f_0, in 5 of 5 repetitions at periods of 7, 137 and 5,000 lags, and at ε_thr in about half, as a confidence bound requires. The analytic threshold was also checked against 400 permutations of the lag ordering: their maxima have mean 9.81 and 95th percentile 12.06, and 2.2% exceed z = 12.90, against the 2.5% per channel that Bonferroni promises.

7 Limitations

1. The exclusion map of §6.3 derives from one pulse (28297); the joint bound of §6.4 uses all ten, as do the setting-correlation test of §5.1 and the single-party memory scan of §6.6, the parity scan of §6.7, and the outcome–outcome scan of §6.5. The injection verification (Figure 4) was run on 28297 alone. α varies across the archive by 20% (standard deviation, against 0.94% measurement error per pulse), tracking the falling click rate. Round 28297 has the third-smallest α of the ten — only rounds 28293 and 28294 lie lower, by 0.8% and 1.6% — and since $\varepsilon_\mathrm{excl} \propto 1/\alpha$ the single-pulse map is within 2% of the loosest that any of the ten would yield: at equal n, another pulse would give a bound 0.60× to 1.02× the stated value. That restriction is therefore conservative to within 2% rather than merely unquantified,

and it is why the joint improvement exceeds $\sqrt{10}$.

2. Oscillatory kernels are covered only in part. The map of §6.3 does not reach them at all: its $1/\sqrt{Q}$ regularity is empirical and established only within the tested class of one-signed kernels, and an oscillatory kernel has a different optimal filter shape entirely. §6.8 covers the *purely sinusoidal* case, where all the power sits at one frequency, and bounds such a coupling of unit peak at $\varepsilon < 5.5 \times 10^{-4}$ across all ten pulses. It does *not* fully cover a modulated or damped oscillation — a decaying sinusoid, or one whose amplitude varies over its support — because an envelope spreads the power over a band of frequencies, so no single periodogram bin carries the whole signal and the sensitivity degrades by roughly the number of bins the power is spread across. Between the two lies the same gap as before: a kernel that is neither one-signed nor a single sinusoid is constrained by neither statistic at full sensitivity.
3. The injection places the coupling in one channel. A model acting coherently in both was not tested.
4. Lags $|k| > 10{,}000$ and widths $\tau > 10{,}000$ were not examined; the window already constrains sensitivity above $\tau \approx 3{,}000$.
5. Time and calibration are confounded across the spread pulses; a difference could not be attributed to either. The conversion from trials to seconds is an upper bound and is not constant across the archive (§4.1).
6. Measurement independence is bounded, not excluded. Models below threshold remain viable, and within-trial retrocausality is entirely untouched.
7. The bounds concern dependence of an outcome on a single setting (§6.1–6.6) or on the parity of two adjacent settings (§6.7), matching the model class of §2, which is linear in the settings. Dependence on functionals of three or more settings, or of non-adjacent pairs, with vanishing lower-order marginals is not constrained.

8 Conclusion

Across ten pulses of a loophole-free Bell test spanning twenty-two months, no correlation was found between measurement outcomes and settings chosen at any other trial. The statement to carry away is the model-independent one: I(O ; S at lag k) $< 1.13 \times 10^{-6}$ bits per trial for every lag up to $\pm 10{,}000$, at family-wise $\alpha = 0.05$ — 1/362 of the same-trial correlation the same instrument registers — while the outcome–outcome information at nonzero lag stays below 1/13,384 of the Bell correlation, and no outcome lets any other trial's setting be guessed with a bias above 1.25×10^{-3} (§6.1). The instrument is not insensitive.

Translated into the parameters of the model class of §2, the same data exclude ε above 7.6×10^{-2} at $\tau = 1$ and 4.4×10^{-4} at $\tau = 10^4$ on a single pulse, across four kernel shapes including the future-directed ones natural to retrocausal models. That map is the bits above re-expressed in model units, not a second result; within the tested class the sensitivity follows an approximately universal $1/\sqrt{Q}$ scaling, and the rigorous exclusion is stated for the four explicit families. Combining the ten pulses by weight, with the pulses shown to be uncorrelated, the excluded coupling falls to 1.9×10^{-2} and 1.2×10^{-4} at the same two widths, and the weakest point of the entire joint map — the narrowest kernel, $\tau = 1$ — still excludes ε above 4.34×10^{-2}. The same holds within each wing separately — neither device shows any correlation with its own settings at nonzero lag, in any of the ten pulses (§6.6) — and for the simplest joint functional of the settings, the parity of adjacent pairs (§6.7). A frequency scan of all ten pulses adds the one shape the matched filters cannot see, an oscillatory kernel, and finds nothing above threshold in 200,000 frequencies (§6.8).

What would a nonzero ε have implied, and what does its absence buy? Within the model class every kernel excludes k = 0 (§2.2) and so never touches the same-trial statistic; cross-trial coupling of any strength cannot by itself fake the observed Bell violation — Figure 2 shows this directly. Its operational cost lies elsewhere: in the assumption, made whenever such records certify randomness, that trials do not leak information about one another. That leakage is what the bound prices — less than 1.13×10^{-6} bits per trial about any single setting up to ten thousand trials away, or a guessing bias below 1.25×10^{-3}. The bound cannot be converted into a bound on Hall's $I(X, Y : \Lambda)$; the data-processing inequality runs the wrong way (§1). What is constrained is the operationally effective part of measurement dependence, not the hidden variable itself — the strongest statement data of this kind permit, and the weakest a skeptic must grant.

This does not refute retrocausal models; the class they occupy is within-trial and is not addressed here. What it does is convert an untested assumption into a measured constraint — including the no-memory assumption on which device-independent randomness and DI-QKD rest, verified here on an operating beacon at the stated sensitivity, without a claim of security. The region is now mapped.

A Which script produces which number

Table 12: Which script produces which number.

Script	Produces
`code/katevasma.py`, `code/load_curby.py`	download and unpack the beacon records; SHA-256 manifest
`code/xartografisi.py`	§4: protocol parameters of 25 rounds (`results/xartografisi_cache.json`)
`code/full_scan.py`	§6.1: 20,001-lag scan, $\chi^2(1)$ calibration, thresholds, Figure 1 data
`code/j_curve.py`	§6.2: J(k), the shift-sign self-test, Figure 2 data
`code/stability10.py`	§6.2: the ten-pulse J values and the 0-of-1,060 count
`code/lag_dense.py`, `code/lag_test.py`	the 111-lag scans of Limitation 1; the deadtime entry of §5
`code/meros1_alpha.py`	§3: r_1, r_2, $\delta(0)$, α, C, exact vs approximate I
`code/meros2_injection.py`	§3: injection test of the analytic relation, clipping
`code/meros3_map.py`, `code/meros3_verify.py`	§6.3: matched filter, symmetric map, injection verification
`code/meros4_outcome_outcome.py`	§6.5: outcome–outcome scan and k = 0 control
`code/meros5_asym.py`, `code/meros5_verify.py`	§6.3: the four kernels, mirror-filter test
`code/meros6_p0.py`	§3: p_0 as marginal, exact MI, the 2.2×10^{-16} reconstruction
`code/meros6_systematic.py`	§6.3: decomposition of the 3–7% systematic
`code/meros6_kernelQ.py`	§6.3: $\varepsilon \propto 1/\sqrt{Q}$ and the Q $\rightarrow \varepsilon_{\rm excl}$ table
`code/meros6_alpha10.py`	Limitation 1: α for each of the ten pulses
`code/meros7_power.py`	§6.3: the $\varepsilon = 0$ and $\varepsilon_{\rm excl}/2$ levels of Figure 4
`code/meros8_settings.py`	§5.1: setting correlations and the phantom signal
`code/meros9_joint.py`	§6.4: per-pulse analysis and the inverse-variance joint bound
`code/meros10_settings10.py`	§5.1: the phantom check across all ten pulses
`code/meros11_optimality.py`	§6.3: Gaussianity of $\hat{\delta}$, kernel-mismatch loss
`code/meros12_selfmemory.py`	§6.6: single-party memory scan, all ten pulses
`code/meros13_parity.py`	§6.7: setting-pair parity scan, all ten pulses
`code/meros14_oo10.py`	§6.5: the ten-pulse outcome–outcome scan
`code/meros15_homogeneity.py`	§6.4: ε-homogeneity test; §6.3: matched-threshold factors
`code/meros16_crosspulse.py`	§6.4: cross-pulse correlation of $\hat{\delta}(k)$, the pulse-independence row of Table 3
`code/meros17_periodogram.py`	§6.8: Lomb–Scargle periodogram of $\hat{\delta}(k)$, its threshold, null and positive control
`code/meros18_covariance.py`	§6.4: covariance of the $T_{\rm p}$ across pulses, per filter, and its permutation null
`code/check_refs.py`	verifies every internal cross-reference resolves
`code/figures_en.py`	Figures 1–3

B Reproduction

```
python3 code/katevasma.py --rounds 28297
python3 code/load_curby.py curby_round_28297.bin --out curby_28297.npz
```

```
python3 code/full_scan.py ; python3 code/meros1_alpha.py
python3 code/meros2_injection.py ; python3 code/meros3_map.py
python3 code/meros3_verify.py ; python3 code/meros4_outcome_outcome.py
python3 code/meros5_asym.py ; python3 code/meros5_verify.py
python3 code/meros6_p0.py ; python3 code/meros6_kernelQ.py
python3 code/meros6_systematic.py ; python3 code/meros6_alpha10.py
python3 code/meros7_power.py ; python3 code/meros8_settings.py
python3 code/meros9_joint.py ; python3 code/meros10_settings10.py
python3 code/meros11_optimality.py ; python3 code/meros12_selfmemory.py
python3 code/meros13_parity.py ; python3 code/meros14_oo10.py
python3 code/meros15_homogeneity.py ; python3 code/meros16_crosspulse.py
python3 code/meros17_periodogram.py ; python3 code/meros18_covariance.py
python3 code/figures_en.py
```

Random seeds are fixed in the scripts: 4711 (exclusion map shuffles), 808 and 909 (injection verification), 1707 (power curve), 2026 (systematic diagnostics), 3141 (mismatch), 5150 (joint bound), 0 and |k|+1 (per-lag nulls). Results were produced with Python 3.14.6, NumPy 2.5.1, SciPy 1.18.0 and Matplotlib 3.11.1; the minimum supported versions are given in `requirements.txt`.

C Derivations

C.1 Second-order expansion of the mutual information

Let the setting take two values with equal probability, and write r_1, r_2 for the conditional click rates, $p_0 = (r_1 + r_2)/2$ for the marginal, and $\delta = (r_2 - r_1)/2$. The mutual information of the 2×2 table is

$$I \;=\; \sum_{s,o} P(s,o)\;\log_2\!\left[\frac{P(s,o)}{P(s)P(o)}\right] \tag{13}$$

With $P(s) = \frac{1}{2}$ and $r_{1,2} = p_0 \mp \delta$, expanding each term to second order in δ/p_0 and $\delta/(1-p_0)$:

$$P(o{=}1|s)\;\log_2\!\left[\frac{P(o{=}1|s)}{p_0}\right] \;=\; (p_0 \pm \delta)\log_2\!\left(1 \pm \frac{\delta}{p_0}\right) \;=\; \frac{\pm\delta + \delta^2/2p_0}{\ln 2} + O(\delta^3) \tag{14}$$

and similarly for o=0 with $p_0 \to 1 - p_0$. Averaging over the two settings, the first-order terms cancel by construction, leaving

$$I \;=\; \frac{1}{\ln 2}\left[\frac{\delta^2}{2p_0} + \frac{\delta^2}{2(1-p_0)}\right] \;=\; \frac{\delta^2}{2\ln 2\; p_0(1-p_0)} + O(\delta^3) \tag{15}$$

The expansion parameter is δ/p_0, not p_0 itself; for the present data $\delta/p_0 = 0.283$ and the residual is 1.35%, in the direction of underestimating I. The exact four-term sum is used wherever precision matters; the expansion serves only to exhibit the ε^2 scaling and to define C.

C.2 The Eberhard statistic at nonzero lag

Write the Eberhard [9] combination as

$$J \;=\; N(o{=}11 \,|\, a_1 b_1) - N(o{=}10 \,|\, a_1 b_2) - N(o{=}01 \,|\, a_2 b_1) - N(o{=}11 \,|\, a_2 b_2) \tag{16}$$

where $N(\cdot|\cdot)$ counts trials with the indicated outcome pair and setting pair, and $J \leq 0$ for any local hidden-variable model.

When outcomes are taken from trial $i + k$ with $k \neq 0$ while settings are taken from trial i, the two are statistically independent. Each of the four setting groups then sees the same outcome distribution, and with $n/4$ trials per group and outcome probabilities P_{11}, P_{10}, P_{01} for the joint outcomes:

$$J \;\longrightarrow\; \frac{n}{4}\,[P_{11} - P_{10} - P_{01} - P_{11}] \;=\; -\frac{n}{4}\,(P_{10} + P_{01}) \tag{17}$$

The two P_{11} terms enter with opposite sign and cancel exactly. What survives is minus the exclusive single-click rate, and one subtlety fixes its size: the two outcomes at trial i+k are taken as a *pair* from a single trial, so their joint distribution retains the full Bell correlation of the source. For round 28297 the coincidence rate is $P_{11} = 2.79 \times 10^{-3}$ — 59 times the 4.8×10^{-5} that independent outcomes would give — and the exclusive singles are $P_{10} = 4.14 \times 10^{-3}$ and $P_{01} = 4.08 \times 10^{-3}$. The identity then predicts J = $-30,800$ and, from the Poisson propagation over the four counts, $\sigma_J = 227$, hence J/$\sigma_J = -135.6$ — against the observed mean of -135.5 (§6.2). The decoupled value of J is therefore large and negative rather than near zero, and the observed level is reproduced quantitatively by the identity alone. The relevant test is not whether J is small at $k \neq 0$ but whether it is ever positive; across ten pulses and 106 nonzero lags each, it is not.

Data and code availability

Analysis code, intermediate results, and figures are available at https://github.com/christosgeorgitzikis47/cross-trial-bell-bound. Raw data are public at random.colorado.edu; per-round URLs and SHA-256 digests are listed in `results/curby_manifest.json`.

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References

[1] B. Hensen et al., *Nature* **526**, 682 (2015).

[2] L. K. Shalm et al., *Phys. Rev. Lett.* **115**, 250402 (2015).

[3] G. A. Kavuri et al., "Traceable random numbers from a non-local quantum advantage", *Nature* (2025), DOI 10.1038/s41586-025-09054-3.

[4] M. J. W. Hall, *Phys. Rev. Lett.* **105**, 250404 (2010).

[5] H. Price, *Time's Arrow and Archimedes' Point*, Oxford University Press (1996).

[6] K. B. Wharton and N. Argaman, "Colloquium: Bell's theorem and locally mediated reformulations of quantum mechanics", *Rev. Mod. Phys.* **92**, 021002 (2020).

[7] M. S. Leifer and M. F. Pusey, *Proc. R. Soc. A* **473**, 20160607 (2017).

[8] R. Chaves et al., "Causal networks and freedom of choice in Bell's theorem", arXiv:2105.05721 (2021).

[9] P. H. Eberhard, *Phys. Rev. A* **47**, R747 (1993).

[10] J. Barrett, D. Collins, L. Hardy, A. Kent, S. Popescu, *Phys. Rev. A* **66**, 042111 (2002).

[11] R. D. Gill, "Accardi contra Bell (cum mundi): the impossible coupling", in *Mathematical Statistics and Applications: Festschrift for Constance van Eeden*, IMS Lecture Notes — Monograph Series **42**, 133 (2003).

[12] Y. Zhang, S. Glancy, E. Knill, *Phys. Rev. A* **84**, 062118 (2011).

[13] P. Bierhorst, *J. Phys. A* **48**, 195302 (2015).

[14] J. E. Pope, A. Kay, *Phys. Rev. A* **88**, 032110 (2013).

[15] G. Pütz, D. Rosset, T. J. Barnea, Y.-C. Liang, N. Gisin, *Phys. Rev. Lett.* **113**, 190402 (2014).

[16] D. Aktas et al., *Phys. Rev. Lett.* **114**, 220404 (2015).

[17] J. Handsteiner et al., *Phys. Rev. Lett.* **118**, 060401 (2017).

[18] D. Rauch et al., *Phys. Rev. Lett.* **121**, 080403 (2018).

[19] J. Barrett, R. Colbeck, A. Kent, *Phys. Rev. Lett.* **110**, 010503 (2013).

[20] K. B. Wharton, "A novel interpretation of the Klein-Gordon equation", *Found. Phys.* **40**, 313 (2010), arXiv:0706.4075.

[21] K. B. Wharton, "Time-symmetric boundary conditions and quantum foundations", *Symmetry* **2**, 272 (2010).

[22] J. Barrett, N. Gisin, "How much measurement independence is needed in order to demonstrate nonlocality?", *Phys. Rev. Lett.* **106**, 100406 (2011), arXiv:1008.3612.

[23] M. J. W. Hall, C. Branciard, "Measurement-dependence cost for Bell nonlocality: Causal versus retrocausal models", *Phys. Rev. A* **102**, 052228 (2020), arXiv:2007.11903.

[24] D. E. Koh, M. J. W. Hall, Setiawan, J. E. Pope, C. Marletto, A. Kay, V. Scarani, A. Ekert, "Effects of reduced measurement independence on Bell-based randomness expansion", *Phys. Rev. Lett.* **109**, 160404 (2012), arXiv:1202.3571.

[25] J.-Å. Larsson, "Loopholes in Bell inequality tests of local realism", *J. Phys. A: Math. Theor.* **47**, 424003 (2014).

[26] E. Santos, "On possible memory effects in tests of Bell inequalities", arXiv:1603.04428 (2016).

[27] M. Weilenmann, C. Budroni, M. Navascués, "Memory attacks in network nonlocality and self-testing", *Quantum* **9**, 1735 (2025).